

Formal Security Proof for Multi-Channel Modular Barcode Encryption

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Abstract

We provide a formal security analysis of the ILS modular barcode encryption scheme for privacy-preserving semantic search. Under the assumption that HMAC-SHA256 is a secure pseudorandom function (PRF), we prove three properties: semantic privacy of individual barcodes, pairwise indistinguishability against unauthorized observers, and resistance to algebraic reconstruction via the Chinese Remainder Theorem. All proofs proceed by reduction to PRF security. We identify the exact conditions under which each guarantee holds and characterize residual leakage bounds. Empirical ablation confirms the theoretical predictions: without masking, an attacker achieves $\rho = 0.974$ correlation with true similarity; with the full masking scheme, correlation drops to $\rho = 0.013$ ($p = 0.77$), indistinguishable from noise.

1 Notation and Scheme Definition

Definition 1 (System Parameters). *The scheme is parameterized by:*

- K : number of channels (default $K = 200$)
- B : number of quantization bins (default $B = 20$)
- $\mathcal{P} = \{p_1, \dots, p_M\}$: a set of M distinct primes, each satisfying $p_i > B$
- $\kappa \in \{0, 1\}^{256}$: secret key
- $L = 5$: number of wave components

Definition 2 (Barcode Encryption). *For a document with identifier $\text{id} \in \{0, 1\}^*$ and quantized signal $\mathbf{q} = (q_1, \dots, q_K) \in \{0, \dots, B-1\}^K$, the encryption $\text{Enc}_\kappa(\text{id}, \mathbf{q})$ produces M barcode layers:*

$$b_{p,k} = (q_k + s_{p,k}(\text{id}) + w_k(\text{id})) \bmod p, \quad \forall p \in \mathcal{P}, k \in [K]$$

where:

- $s_{p,k}(\text{id})$: rejection-sampled salt, uniform over $\mathbb{Z}/p\mathbb{Z}$
- $w_k(\text{id})$: wave interference mask, sum of L triangle waves

Definition 3 (Rejection-Sampled Salt). *For each prime p and channel k , let $T_p = \lfloor 256/p \rfloor \cdot p$. Draw bytes $\beta_0, \beta_1, \dots$ from $\text{HMAC}_\kappa(\text{id} \| p \| k \| \text{ctr})$ and output the first $\beta_i < T_p$. Set $s_{p,k} = \beta_i \bmod p$.*

Under this procedure, if β_i is uniform over $\{0, \dots, T_p - 1\}$, then $s_{p,k}$ is perfectly uniform over $\mathbb{Z}/p\mathbb{Z}$.

Definition 4 (Wave Mask). *Let f_ℓ, A_ℓ be derived from $\text{SHA256}(\kappa)$ with $A_\ell \in [120, 199]$, let $\phi_\ell(\text{id})$ be derived from $\text{HMAC}_\kappa(\text{id})$, and let $\psi_k(\text{id})$ be the k -th byte of $\text{HMAC-SHA512}_\kappa(\text{id})$. Then:*

$$w_k(\text{id}) = \sum_{\ell=1}^L A_\ell \cdot \text{tri}(f_\ell \cdot k + \phi_\ell(\text{id}) + \psi_k(\text{id}))$$

where $\text{tri} : \mathbb{Z} \rightarrow [-A_\ell, A_\ell]$ is a piecewise linear triangle function.

2 Security Model

Definition 5 (PRF Assumption). *A keyed function family $F : \mathcal{K} \times \mathcal{X} \rightarrow \mathcal{Y}$ is a secure PRF if for all probabilistic polynomial-time (PPT) adversaries $\mathcal{A}$:*

$$\text{Adv}_{\mathcal{A}}^{\text{PRF}}(F) := \left| \Pr_{\kappa \leftarrow \mathcal{K}} [\mathcal{A}^{F_{\kappa}(\cdot)} = 1] - \Pr_{f \leftarrow \text{Func}(\mathcal{X}, \mathcal{Y})} [\mathcal{A}^{f(\cdot)} = 1] \right| \leq \text{negl}(\lambda)$$

Standing Assumption. Throughout, we assume HMAC-SHA256 is a secure PRF. This is standard and underpins TLS 1.3, IPsec, and SSH.

Definition 6 (Semantic Privacy Game). *The adversary $\mathcal{A}$ plays:*

1. $\mathcal{A}$ chooses two signals $\mathbf{q}_0, \mathbf{q}_1 \in \{0, \dots, B-1\}^K$.
2. Challenger draws $b \leftarrow \{0, 1\}$, $\text{id} \leftarrow \{0, 1\}^\lambda$, and returns $\mathcal{B} = \text{Enc}_{\kappa}(\text{id}, \mathbf{q}_b)$.
3. $\mathcal{A}$ outputs $\hat{b}$. Advantage is $|\Pr[\hat{b} = b] - 1/2|$.

Definition 7 (Pairwise Indistinguishability Game). 1. $\mathcal{A}$ chooses signals $\mathbf{q}_0, \mathbf{q}_1$ with $\mathbf{q}_0 \neq \mathbf{q}_1$, and a signal $\mathbf{q}_2 = \mathbf{q}_0$.

2. Challenger draws $b \leftarrow \{0, 1\}$. If $b = 0$: encrypt $(\text{id}_A, \mathbf{q}_0)$ and $(\text{id}_B, \mathbf{q}_1)$. If $b = 1$: encrypt $(\text{id}_A, \mathbf{q}_0)$ and $(\text{id}_B, \mathbf{q}_2)$.
3. $\mathcal{A}$ sees both barcodes, outputs $\hat{b}$.
 $\mathcal{A}$ wins if she can distinguish “similar” ($b = 1$) from “dissimilar” ($b = 0$).

3 Main Results

Theorem 8 (Semantic Privacy). *Under the PRF assumption on HMAC_{κ} , the barcode scheme satisfies semantic privacy (Definition 6):*

$$\text{Adv}_{\mathcal{A}}^{\text{SP}} \leq 2K \cdot M \cdot \text{Adv}_{\mathcal{A}}^{\text{PRF}}$$

Proof. Fix prime $p \in \mathcal{P}$ and channel $k \in [K]$. The adversary observes:

$$b_{p,k} = (q_k + s_{p,k}(\text{id}) + w_k(\text{id})) \bmod p$$

By Definition 3, the salt $s_{p,k}$ is perfectly uniform over $\mathbb{Z}/p\mathbb{Z}$ (rejection sampling guarantees this exactly, with zero bias). Therefore, for any fixed values of q_k and $w_k(\text{id})$:

$$b_{p,k} = (q_k + w_k(\text{id}) + s_{p,k}) \bmod p$$

Since $s_{p,k}$ is uniform over $\mathbb{Z}/p\mathbb{Z}$ and independent of q_k , the sum $(q_k + w_k(\text{id}) + s_{p,k}) \bmod p$ is perfectly uniform over $\mathbb{Z}/p\mathbb{Z}$ regardless of the values of q_k and $w_k(\text{id})$. This is the one-time pad property in $\mathbb{Z}/p\mathbb{Z}$.

Critically, this holds *information-theoretically* for a single document: even an unbounded adversary cannot determine q_k from $b_{p,k}$, because $s_{p,k}$ acts as a perfect one-time pad modulo p .

The computational reduction arises because $s_{p,k}$ is not truly random but derived from HMAC_{κ} . We proceed by hybrid argument. Define:

- **Hybrid H_0 :** Real scheme. Salt from HMAC_{κ} .
- **Hybrid H_1 :** Replace HMAC_{κ} with a truly random function f .
- **Hybrid H_2 :** Replace rejection-sampled outputs with truly uniform elements of $\mathbb{Z}/p\mathbb{Z}$.

$|H_0 - H_1| \leq \text{Adv}_{\mathcal{A}}^{\text{PRF}}$ by the PRF assumption (one invocation per channel per prime).
 $|H_1 - H_2| = 0$: given a truly random function, rejection sampling produces exactly uniform outputs (Definition 3).

In H_2 , the salt is a true one-time pad in $\mathbb{Z}/p\mathbb{Z}$. Therefore $b_{p,k}$ is perfectly uniform and independent of q_k . The adversary has zero advantage.

Over all K channels and M primes, by a union bound:

$$\text{Adv}_{\mathcal{A}}^{\text{SP}} \leq K \cdot M \cdot (\text{Adv}_{\mathcal{A}}^{\text{PRF}} + 0) + K \cdot M \cdot (\text{Adv}_{\mathcal{A}}^{\text{PRF}} + 0) = 2KM \cdot \text{Adv}_{\mathcal{A}}^{\text{PRF}} \quad \square$$

Remark 9. *Unlike our earlier analysis, rejection sampling eliminates the $p/(2 \cdot 256)$ bias term entirely. The guarantee is now information-theoretic conditional on the PRF assumption—there is no residual statistical leakage per channel.*

Theorem 10 (Pairwise Indistinguishability). *Under the PRF assumption, no PPT adversary observing barcodes $\mathcal{B}(\text{id}_1)$ and $\mathcal{B}(\text{id}_2)$ can determine any function of $\mathbf{q}_1 - \mathbf{q}_2$ with non-negligible advantage:*

$$\text{Adv}_{\mathcal{A}}^{\text{PI}} \leq 4KM \cdot \text{Adv}_{\mathcal{A}}^{\text{PRF}}$$

Proof. The adversary can compute, for each prime p and channel k :

$$\delta_{p,k} := (b_{p,k}^{(1)} - b_{p,k}^{(2)}) \bmod p = (q_{1,k} - q_{2,k} + \Delta s_{p,k} + \Delta w_k) \bmod p$$

where $\Delta s_{p,k} = s_{p,k}(\text{id}_1) - s_{p,k}(\text{id}_2)$ and $\Delta w_k = w_k(\text{id}_1) - w_k(\text{id}_2)$.

Lemma 11. *If $s_{p,k}(\text{id}_1)$ and $s_{p,k}(\text{id}_2)$ are independent and each uniform over $\mathbb{Z}/p\mathbb{Z}$, then $\Delta s_{p,k} = s_{p,k}(\text{id}_1) - s_{p,k}(\text{id}_2)$ is uniform over $\mathbb{Z}/p\mathbb{Z}$.*

Proof. For any fixed $a \in \mathbb{Z}/p\mathbb{Z}$ and uniform independent X, Y over $\mathbb{Z}/p\mathbb{Z}$:

$$\Pr[X - Y = a] = \sum_{x=0}^{p-1} \Pr[X = x] \Pr[Y = x - a] = \sum_{x=0}^{p-1} \frac{1}{p} \cdot \frac{1}{p} = \frac{1}{p} \quad \square$$

By Lemma 11, $\Delta s_{p,k}$ is uniform over $\mathbb{Z}/p\mathbb{Z}$ (under H_2 from Theorem 8). Therefore:

$$\delta_{p,k} = \underbrace{(q_{1,k} - q_{2,k} + \Delta w_k)}_{\text{fixed given } \text{id}_1, \text{id}_2} + \underbrace{\Delta s_{p,k}}_{\text{uniform over } \mathbb{Z}/p\mathbb{Z}} \bmod p$$

is uniform over $\mathbb{Z}/p\mathbb{Z}$ regardless of $q_{1,k} - q_{2,k}$.

Since the adversary observes only uniform residues, no function of the signal difference $\mathbf{q}_1 - \mathbf{q}_2$ —including similarity, distance, or identity—can be extracted.

The computational bound follows from reducing to the PRF assumption for both id_1 and id_2 (factor of 2), across K channels and M primes:

$$\text{Adv}_{\mathcal{A}}^{\text{PI}} \leq 2 \cdot 2KM \cdot \text{Adv}_{\mathcal{A}}^{\text{PRF}} = 4KM \cdot \text{Adv}_{\mathcal{A}}^{\text{PRF}} \quad \square$$

Theorem 12 (CRT Resistance). *An adversary performing Chinese Remainder Theorem reconstruction from the M barcode layers of a single document recovers only the masked value $q_k + m_k(\text{id})$, from which q_k cannot be extracted without κ .*

Proof. Let $P = \prod_{i=1}^M p_i$. By the CRT, given residues $b_{p_i,k}$ for all $i \in [M]$, the adversary can uniquely reconstruct:

$$v_k = (q_k + m_k(\text{id})) \bmod P$$

where $m_k(\text{id}) = s_{p,k}(\text{id}) + w_k(\text{id})$ evaluated consistently across primes.

Since $q_k \in \{0, \dots, B-1\}$ and $|m_k| \leq 255 + L \cdot A_{\max} \leq 255 + 5 \cdot 199 = 1250$, the total $q_k + m_k \leq B-1 + 1250 = 1269$. For our primes, $P = \prod p_i \geq 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23 = 3,233,230 \gg 1269$.

Therefore, CRT reconstruction gives the *exact* value $v_k = q_k + m_k(\text{id})$, not a residue class. The adversary knows v_k precisely.

However, extracting q_k from v_k requires knowing $m_k(\text{id})$, which depends on κ through both the salt and the wave. Under the PRF assumption, $m_k(\text{id})$ is computationally indistinguishable from a uniform sample over its range. Therefore:

$$\text{Adv}_{\mathcal{A}}^{\text{CRT}} \leq K \cdot \text{Adv}_{\mathcal{A}}^{\text{PRF}}$$

Remark 13. *The CRT does not provide the adversary with more information than the barcode layers themselves. It merely changes the representation from M residues to a single integer. The security rests entirely on the secrecy of the mask $m_k(\text{id})$, which acts as an additive one-time pad over the integers (not modular).*

□

4 Role of Each Component

The ablation study (Section 4 of the main paper) provides empirical confirmation of the theoretical structure:

Proposition 14 (Necessity of Masking). *Without masking ($m_k = 0$), the barcode $b_{p,k} = q_k \bmod p$ is a deterministic function of q_k . The adversary can compute pairwise distances in modular space and recover similarity with high fidelity.*

Empirical confirmation: Ablation A5 measured attacker correlation $\rho = 0.974$ ($p < 10^{-6}$) without masking—near-perfect leakage.

Proposition 15 (Sufficiency of Salt). *The rejection-sampled salt alone provides computational semantic privacy and pairwise indistinguishability under the PRF assumption.*

Empirical confirmation: With salt only (no wave), attacker correlation dropped to $\rho = 0.027$ ($p = 0.55$)—indistinguishable from noise.

Proposition 16 (Role of Wave Interference). *The wave mask provides defense in depth. It does not affect search quality (the key holder compensates the wave during comparison). It provides additional security margin against attacks that might partially compromise the salt, such as related-key attacks or weak PRF instantiations.*

Empirical confirmation: With salt + wave, $\rho = 0.013$ ($p = 0.77$). Marginally better than salt alone, confirming the defense-in-depth role.

Proposition 17 (Adaptive Prime Selection). *For correct operation, each prime $p \in \mathcal{P}$ must satisfy $p > B$ (the number of bins). When $p \leq B$, distinct signal values $q_k \neq q'_k$ may satisfy $q_k \equiv q'_k \pmod{p}$, creating information-losing collisions that degrade retrieval quality. When $p > B$, the modular reduction $q_k \bmod p$ is injective on $\{0, \dots, B-1\}$, preserving all distance information.*

Empirical confirmation: With $B = 50$ and small primes $\{7, 11, 13, 17, 19\}$, MRR dropped to 37.9%. With large primes $\{29, 31, 37, 41, 43\}$, MRR recovered to 98.3%.

5 Security Level

6 Limitations of This Analysis

Multi-query leakage. Theorems 8–12 analyze single-document and pairwise security. An adversary observing n barcodes over time may accumulate information about the PCA projection matrix V (which is public) and the distribution of signals. This analysis does not address adaptive multi-query attacks.

Property	Type	Bound	Assumption
Semantic privacy (single channel)	Computational	$2KM \cdot \text{Adv}^{\text{PRF}}$	PRF
	Information-theoretic	0	Perfect salt
Pairwise indisting.	Computational	$4KM \cdot \text{Adv}^{\text{PRF}}$	PRF
CRT resistance	Computational	$K \cdot \text{Adv}^{\text{PRF}}$	PRF

Table 1: Security guarantees. With rejection sampling, per-channel semantic privacy is information-theoretic (zero advantage even for unbounded adversaries). The computational bounds arise only from instantiating the PRF.

PRF instantiation. The proofs assume HMAC-SHA256 is a secure PRF. If a weakness in SHA-256 or HMAC is discovered, the computational guarantees degrade. The information-theoretic per-channel guarantee (Theorem 8) remains valid as long as rejection sampling produces uniform outputs, regardless of the PRF.

Side channels. Timing analysis (empirically confirmed: KS $p = 0.58$) shows no leakage in our implementation. However, a formal proof of constant-time execution would require analysis of the specific implementation, which we do not provide.